

Deep-Dive Study Guide — Index

AI-Powered Financial Document Similarity Finder

Each file is a standalone deep dive into one technology: every component explained, the why behind each design choice, how it's used in this project, common pitfalls, practice exercises, and self-check questions. Study them one at a time.

Modules

1. [PostgreSQL — Relational Database](#)
 2. [Qdrant — Vector Database](#)
 3. [React — Frontend Framework](#)
 4. [Spring Boot — Java Backend](#)
 5. [Python + Sentence Transformers — Embedding Service](#)
 6. [Docker & Docker Compose — Infrastructure](#)
 7. [Tesseract OCR — Text from Images](#)
 8. [JWT — Authentication & Authorization](#)
 9. [Apache PDFBox & Apache POI — File Parsing](#)
 10. [How Everything Connects Together](#)
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Recommended study order

- 1 PostgreSQL → every component stores/retrieves data here
- 2 Docker → you need it to run Qdrant + PostgreSQL locally
- 3 Python/Flask → simplest intro to REST APIs
- 4 Sentence Transformers → vectors are the core concept of the project
- 5 Qdrant → now you know what you're storing and why
- 6 Spring Boot → the orchestrator that connects everything
- 7 React → the user-facing layer calling the API
- 8 JWT → security wraps everything; easier after the APIs
- 9 PDFBox/POI/Tesseract → the input side: how data enters the pipeline
- 10 Everything Together → trace one upload and one search end to end

Tip: read Module 10 last, but skim it first too — seeing the whole map makes each individual module easier to place.

Covers: PostgreSQL · Qdrant · React · Spring Boot · Python/Sentence Transformers · Docker · Tesseract OCR · JWT · Apache PDFBox · Apache POI